Can Vision-Language-Action Models Adapt to Visual Perturbations Using a Test-Time Trajectory?

Jiho Lee, Heayoun Choi, and Eunwoo Kim

***Abstract*—Recently, vision–language–action models (VLAs) have shown strong generalization across diverse robotic tasks. However, when deployed in new environments with unseen visual conditions, their performance often degrades due to visual distribution shifts. Such shifts disrupt the alignment between visual and action representations, leading to distorted action semantics at test-time. Existing approaches primarily address this issue by scaling multi-modal training data or incorporating reasoning mechanisms. However, they require substantial data collection and computational resources, limiting their practicality for various real-world deployments. To address this challenge, we propose *TTA-VLA*, a simple yet effective test-time adaptation framework for VLAs using only a test-time trajectory. Our method leverages environment-specific observations from this trajectory as self-supervision signals for adaptation. Based on these observations, we extract visual and action embedding transitions from a pre-trained VLA and align them to reduce visual–action temporal inconsistencies induced by unseen visual shifts. We further employ the task-language representation as an auxiliary semantic regularizer to encourage task-consistent visual–action transition features. Across various visual perturbation scenarios on benchmark and real-world, our method achieves an average success rate improvement of 16% over the pre-trained policies.**

***Index Terms*—Machine learning for robot control, vision-language-action models, test-time adaptation.**

I. Introduction

VISION-language-action models (VLAs) have shown remarkable progress in enabling robots to interpret natural language instructions from humans, understand complex visual scenes, and execute manipulation behaviors [1], [2], [3], [4]. They jointly learn from large-scale multimodal robotic datasets [5], [6], [7], including images, language instructions, robot sensory data, and robot demonstrations. As a result, VLAs can integrate the capability of perception, reasoning, and action generation into an end-to-end policy, highlighting their potential as a scalable and flexible foundation for generalist robot learning [8], [9].

Despite their success, VLAs still suffer from critical limitations when deployed in new environments [2], [3], [11]. In real world, visual perturbations such as background texture, lighting condition, and camera viewpoints can readily arise and deviate from the training distribution, as illustrated in Fig. 1. The study in [10] shows that existing VLAs are highly sensitive to these perturbations, with average success rates dropping from around 95% to below 30%. Consistently, we observe in our environment that even minor appearance shifts can severely degrade policy performance, yielding an average 70% drop across the six perturbation categories, as in Fig. 1(h).

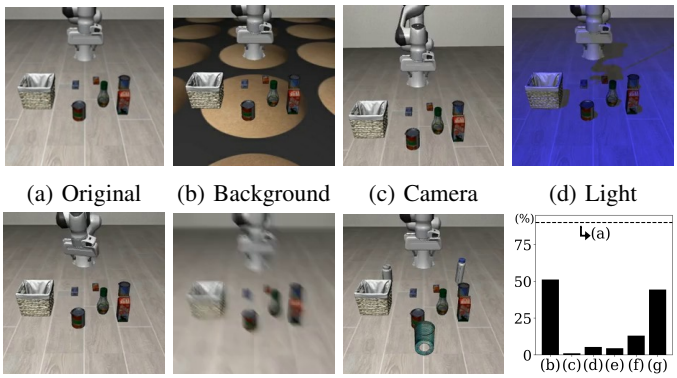


Fig. 1: Examples of various visual perturbation types in LIBERO-Plus [10], along with their performance on OpenVLA [3]. Original performance without visual perturbation, exemplified by (a), is plotted as a dashed line in (h).

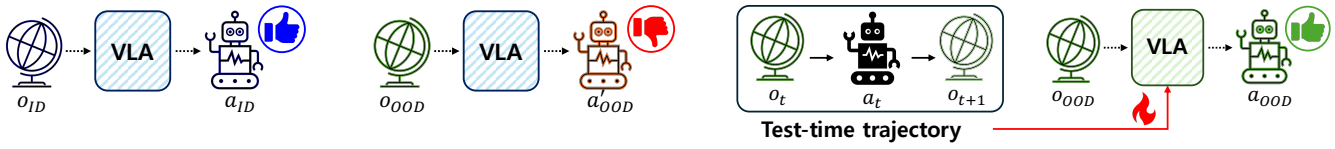
To address these problems, most recent works have mainly focused on advancing the training recipe. A common direction is to increase the scale and diversity of demonstrations collected across varied scenes, objects, and sensing conditions [2], [3], [11], [12]. A broad range of observations and manipulation behaviors helps the model to learn perception–action relationships that generalize more consistently in novel environments. Another direction is to enhance the reasoning capability of the underlying vision-language backbone [4], [13], [14], [15]. Reasoning mechanisms enable VLAs to generate grounded intermediate representations, such as object relations and subtask decomposition, allowing them to leverage the semantic and compositional capabilities of pre-trained large-language models and visual-language models.

While these two directions offer meaningful benefits, they aim to improve the capability of VLAs through extensive training, resulting in substantial additional data collection and computational resources. Furthermore, these advances typically strengthen visual understanding, rather than explicitly addressing distribution shifts at deployment. Consequently, real-world environments vary widely and may include conditions

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(a) In-distribution environment

(b) Out-of-distribution environment (existing works)

(c) Out-of-distribution environment with test-time trajectory-based adaptation (ours)

Fig. 2: Comparison of VLAs under in-distribution and out-of-distribution. Blue denotes the in-distribution training domain, green indicates the out-of-distribution domain at test time, and orange represents inconsistent action outputs induced by distribution shifts. (a) In an in-distribution setting, observations align with the training distribution of the VLA, resulting in consistent visual–action representations. (b) Under an out-of-distribution environment, visually shifted observations lead to mismatches between visual inputs and action representations in the pre-trained VLA. (c) Our method effectively adapts the pre-trained VLA using a test-time trajectory in the shifted environment, mitigating visual–action inconsistencies.

not covered during training, which can lead to inconsistent action behaviors under the resulting distribution shifts, as illustrated in Fig. 2(a) and Fig. 2(b). These considerations motivate investigating complementary approaches that enable models to adapt using information available at test time, allowing more reliable operation under unseen conditions.

To adapt the pre-trained model during deployment without extensive training, test-time adaptation (TTA) provides a natural solution across various domains [16], [17], [18]. It adjusts the model parameters using only the inputs available at test time, typically optimizing self-supervised objectives that encourage confident predictions with entropy minimization, consistency regularization, or pseudo-labeling [19], [20], [21], [22]. Despite its potential benefits, applying test-time adaptation to VLAs poses additional challenges due to their multi-modal characteristics. In particular, VLAs depend on tight alignment across vision, language, and action modalities. As a result, even a modest visual distribution shift can easily disrupt this alignment and compromise downstream action semantics. However, prior TTA approaches in robotics do not consider the alignment of multi-modal representations, only relying on feedback or entropy of predictions as output-level supervision signals [16], [23]. Furthermore, addressing such cross-modal misalignment at test time requires supervision signals at the representation-level that reflect whether actions remain semantically grounded in both the instruction and the scene. Yet, although robots can obtain rich raw sensory observations at deployment, it remains challenging to derive test-time supervision that explicitly captures cross-modal semantic grounding among vision, language, and action, since existing TTA objectives are largely agnostic to these semantics.

To overcome these challenges, we introduce a **T**est-time **T**rajectory-based **A**daptation framework for **VLA**s, named **TTA-VLA**. It exploits visual observations collected from a trajectory in the new deployment environments to explicitly align visual and action representations, as shown in Fig. 2(c). The key insight behind TTA-VLA is that during training, VLAs learn relationships between input observations and output actions, yielding transition patterns in which changes in visual observations induce corresponding action transitions over time. We leverage this temporal consistency between visual and action transitions in their latent representations as a self-supervised signal for test-time adaptation. Under visual domain shifts, however, the pre-trained VLA may fail to preserve its original observation–action relationship, leading to reduced visual–action temporal consistency such that visual changes no longer induce appropriate action transitions. To address this, we extract transition representations from consecutive visual and action embeddings along a test-time trajectory as environment-specific information. Then, we align these transition representations with the task instruction as an auxiliary semantic reference, encouraging the adapted visual–action features to remain consistent with the intended task. In this way, TTA-VLA provides a simple yet effective test-time adaptation framework that restores disrupted visual–action alignment while maintaining semantic stability under distribution shifts, without expert supervision and extensive computational overhead. This formulation is not tailored to a specific VLA architecture, but instead addresses test-time adaptation as a general challenge arising from cross-modal misalignment between visual and action representations.

We validate the effectiveness of our test-time adaptation framework under a wide range of visual distribution shifts on three representative VLAs: OpenVLA [3], OpenVLA-OFT [24], and π_0 [2]. For simulation experiments on LIBERO-Plus [10] with six visual perturbation types, we observe consistent performance improvements over the pre-trained policies, with an average gain of 12%. To assess the real-world practicality, we evaluate it under four visual perturbations, achieving an average gain of 24%.

II. Related Work

A. Vision-language-action models

Built upon the strong compositionality and semantic grounding capabilities of pre-trained vision-language models (VLMs) [25], [26], vision-language-action models (VLAs) achieve powerful generalist robot control by taking sensory observations and language instructions as input and producing low-level actions as output [8], [9]. Recent VLAs adopt diverse action-generation architectures, which can be broadly categorized into three groups: (i) autoregressive policies that extend VLMs to sequential action generation [3], [27], [28], (ii) diffusion-based methods, which produce continuous actions

through iterative denoising [1], [2], [29], [30], and (iii) CLIP-based approach using similarity between visual context and language-based motions [11]. To further strengthen zero-shot generalization, prior works primarily expand the coverage and diversity of multi-modal robotic training datasets, enabling VLAs to generalize across various situations [2], [3], [11], [12]. More recently, several methods augment VLAs with high-level reasoning mechanisms, such as subtask-conditioned action generation [4], [13], [14], [15].

However, they primarily focus on improving the robustness of VLAs through large-scale training with almost 1 million demonstrations [6] or additional reasoning data [15], requiring extensive computational resources. Furthermore, real-world deployments may expose models to conditions beyond the training distribution due to its diversity, inducing distribution shifts that can degrade performance at test-time. To address this, our work focuses on test-time adaptation, explicitly re-aligning disrupted relationships of observation and action representations through a test-time trajectory.

B. Test-time adaptation

Test-time adaptation (TTA) aims to mitigate distribution shift by adjusting the model parameters at test-time using only unlabeled test-time data, typically by optimizing self-supervised objectives that encourage consistent and confident predictions, thereby avoiding costly retraining or additional supervision [31], [32]. TTA has been widely investigated for maintaining feature consistency under distribution shift with entropy minimization [18], [21] or batch-norm statistics adaptation [22], [33]. For vision–language models, prior works explore optimizing text prompts using augmented images [17], [19] or employing self-supervised proxy tasks [20], [34]. In robotics, only a few studies have explored TTA, primarily in visual-language navigation, using entropy minimization of action predictions [23] or binary feedback–based reinforcement learning [16], which exploit output-level signals rather than the internal latent representations of the model. However, TTA for vision–language–action models remains unexplored, despite their susceptibility to visual distribution shifts.

In this work, we introduce test-time adaptation of VLAs by leveraging a test-time trajectory to self-supervise the alignment between visual and action representations disrupted by visual distribution shifts. To stabilize this alignment with task semantics, we incorporate the task instruction as a language-conditioned reference, encouraging visual and action features to remain semantically consistent at the feature-level. In VLAs, test-time scaling has been studied to improve action generation at test-time by producing multiple candidates followed by post-hoc selection [35], which is orthogonal to our focus because it only changes the inference procedure rather than adapting model representations disrupted by distribution shifts.

III. Proposed method

A. Problem statement: test-time adaptation of VLAs

Our goal is to enable vision-language-action models (VLAs) to maintain reliable action generation under visual distribution shifts by adapting to new environments using only a test-time trajectory. Typically, VLAs, initialized from pre-trained VLMs, enable general-purpose robot control by taking sensory observations and a task instruction as input and producing low-level actions as output. VLAs are trained on a large-scale dataset of expert demonstrations $\mathcal{D}_{train} = \{(o_t, i_t, a_t)\}_{t=1}^{N}$, where o_t, i_t, and a_t at time step t denote sensory observation, natural-language task instruction, and its corresponding expert action, respectively. Note that the observation is an RGB image, following a standard VLA setup [3]. The image is first encoded into a visual feature through the visual encoder $\mathcal{F}_v$ as $\overline{z}_t = \mathcal{F}_v(o_t)$ and then transformed by a visual projector $\mathcal{V}$ to make it compatible with the language model, $z_t = \mathcal{V}(\overline{z}_t)$. The task instruction is also embedded using a text encoder $\mathcal{F}_l$ as $e_t = \mathcal{F}_l(i_t)$. The projected visual feature z_t and language feature e_t are concatenated and fed into the language model, which produces a latent representation u_t that is mapped to the robotic action space.[1] In this way, the VLA model parameterized with θ learns the observation–action relationships by minimizing the supervised objective with the action prediction loss $\mathcal{L}_\theta$:

$$\min_{\theta} \mathbb{E}_{(o_t,\, i_t,\, a_t)\sim\mathcal{D}_{train}} \Big[\mathcal{L}_\theta\big(a_t \mid o_t, i_t\big)\Big]. \tag{1}$$

When VLAs are deployed in a new environment, they encounter new sensory observations in $\mathcal{D}_{new}$ that differ from the distribution of training data $\mathcal{D}_{train}$ due to changes in background, lighting, viewpoint, or object appearance. In the context of VLAs, such visual domain shifts do not merely distort visual features as typically in vision tasks, they fundamentally disrupt the tri-modal alignment between vision, language, and action. Specifically, the visual embedding z_t becomes misaligned from the joint embedding space where language instructions e_t and action sequences a_t are anchored, leading to action predictions that are no longer semantically grounded. Therefore, we define the primary challenge of TTA for VLAs as adapting visual representations to the new environment while preserving the cross-modal semantic consistency.

At test-time, however, VLAs no longer have access to expert actions, ground-truth labels, or additional demonstrations that could be used to recover this misalignment. The only available signal is incoming sensory observations from the new environment, which may significantly differ from the training distribution. Consequently, we formulate the TTA problem as the following unsupervised optimization:

$$\theta = \underset{\theta_o}{\operatorname{argmin}}\, \mathbb{E}_{(o_t,\, i_t)\sim\mathcal{D}_{new}} \big[\mathcal{L}^{\text{TTA}}_{\theta_o}\big(o_t, i_t, \theta_o\big)\big], \tag{2}$$

where $\mathcal{L}^{\text{TTA}}_{\theta_o}$ denotes an objective function for test-time adaptation, and θ_o represents the pre-trained model parameters. This adjusts the pre-trained parameters by leveraging its behavior when exposed to observations from a new distribution.

[1]For example, in OpenVLA, z_t and u_t are obtained from the visual projector output and the final LLM hidden states, respectively; in π_0, they are obtained from the PaliGemma visual projector output and the final PaliGemma language model hidden states.

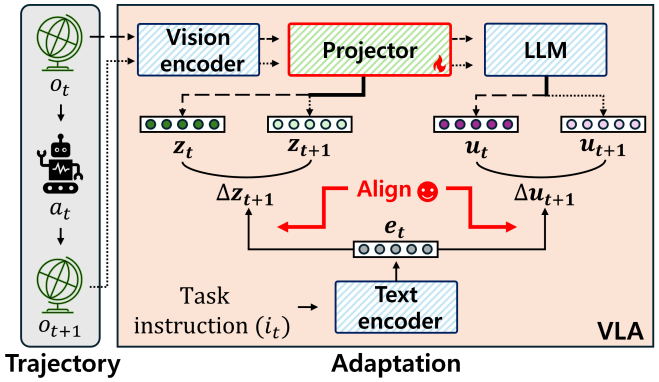


Fig. 3: Overview. The robot collects observations by executing a trajectory in the new environment. Each observation is converted into visual and action features, z and u, through the pre-trained VLA. Our proposal aligns their transitions, Δz_{t+1} and Δu_{t+1}, while anchoring this alignment to the task-instruction embedding e_t to ensure that both transitions remain semantically consistent with the task i_t.

B. TTA-VLA: Test-time Trajectory-based Adaptation for VLAs

To address the above-mentioned challenges in tri-modal misalignment, we propose TTA-VLA, which leverages the temporal structure of observation–action relationships conditioned on language as a self-supervised signal. Specifically, we obtain this signal from a test-time robot trajectory collected in the new environment by arbitrarily executing the robot, from which we capture how the new environment affects the pre-trained behaviors of the VLA, as depicted in Fig. 3. Formally, this trajectory can be written as a sequence of observation–action pairs, $(o_t, a_t, o_{t+1}, a_{t+1}, \ldots)$. From this trajectory, we encode the observations to extract visual features from the visual projector, z_t and z_{t+1}, and action features from the language model, u_t and u_{t+1}, with the same hidden dimensionality. We then compute their temporal transitions as follows:

$$\Delta z_{t+1} = z_{t+1} - z_t \quad \text{and} \quad \Delta u_{t+1} = u_{t+1} - u_t. \tag{3}$$

These token-wise transitions are then mean-pooled over tokens. Our key insight for feature alignment is that VLAs implicitly learn a temporal correspondence between changes in observations and the action representations during training. However, visual domain shifts disrupt this correspondence, motivating the restoration of consistency between the visual transition Δz_{t+1} and the action transition Δu_{t+1} for maintaining reliable behavior of VLAs in the new environment.

To this end, we introduce a transition-based adaptation loss as $\mathcal{L}_{\text{align}}(\Delta z_{t+1}, \Delta u_{t+1})$ to reduce the temporal inconsistency between visual and action feature transitions. We investigate two geometric alignment strategies, angular-based and projection-based, that encourage directional consistency between transitions across modality spaces. The angular-based alignment focuses on matching the orientation of feature transitions by minimizing their cosine distance, inspired by [36]:

$$1 - \frac{\Delta z_{t+1}^{\top} \Delta u_{t+1}}{|\Delta z_{t+1}|_2 |\Delta u_{t+1}|_2}. \tag{4}$$

The projection-based variant projects the action feature transition onto the visual feature transition and penalizes the residual, encouraging their geometric consistency, inspired by [37], with a constant for numerical stability ϵ:

$$\left\| \Delta u_{t+1} - \frac{\Delta u_{t+1}^{\top} \Delta z_{t+1}}{\|\Delta z_{t+1}\|_2^2 + \epsilon} \Delta z_{t+1} \right\|_2^2. \tag{5}$$

Their corresponding analyses are provided in Section IV-A2.

As an auxiliary task-semantic regularizer, we use the natural-language task description as an environment-invariant, language-conditioned reference during adaptation. Since the task instruction remains unchanged across visual domains, its embedding provides stable task-level context that helps mitigate semantic drift. Specifically, we incorporate the task instruction as a language semantic anchor by re-centering the visual and action transitions around the text embedding. This gives the following transitions:

$$\Delta z_{t+1}^{anch} = \Delta z_{t+1} - e_t \quad \text{and} \quad \Delta u_{t+1}^{anch} = \Delta u_{t+1} - e_t, \tag{6}$$

where e_t denotes the text embedding of the task instruction i_t extracted from the text encoder $\mathcal{F}_l$. Then, we rewrite the objective function as follows:

$$\mathcal{L}_{\theta_o}^{\text{TTA}} = \mathcal{L}_{\text{align}}(\Delta z_{t+1}^{anch}, \Delta u_{t+1}^{anch}). \tag{7}$$

By incorporating task-semantic information, the proposed method not only helps stabilize adaptation but also mitigates semantic drift, enabling more reliable behavior under visual domain shifts. During adaptation, only the parameters of the visual projector $\mathcal{V}$ are updated, while all other components of the VLA policy remain frozen, following [38], [39]. Accordingly, both transitions change with the adapted visual representation during optimization.

IV. Experiments

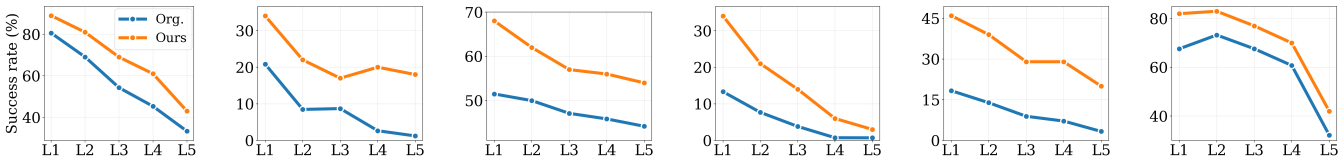
A. Benchmark experiments

1) Setup: Dataset. We validate our proposal on LIBERO-Plus [10], which extends LIBERO [40] with systematic and realistic visual perturbations for evaluating test-time adaptation under visual distribution shifts. We assess performance across six axes of visual domain shifts, including background texture, camera viewpoint, lighting condition, robot initial state, sensor noise, and object layout, with examples shown in Fig. 1. These perturbations expand the original set of 10 tasks to approximately 353 tasks on average across the object, spatial, and goal variants of the benchmark.

Implementation details. We evaluate our method on OpenVLA [3], OpenVLA-OFT [24], and π_0 [2]. We compare our method against the corresponding pretrained VLA without adaptation under two conditions: (i) in-distribution settings without visual perturbations and (ii) out-of-distribution settings with visual perturbations, to assess both the robustness of pretrained VLAs to visual distribution shifts and the effectiveness of our adaptation method. We additionally compare with entropy minimization [18], [21], a standard test-time

TABLE I: Comparison results of task success rates on LIBERO-Plus across six visual perturbation scenarios.

Evaluation setting	Method	Background texture	Camera viewpoint	Light condition	Robot initial state	Sensor noise	Object layout	Avg.
OpenVLA								
In-distribution	Original Policy			90.0%				90.0%
Out-of-distribution	Original Policy	46.4%	1.2%	11.4%	3.7%	5.9%	46.5%	19.2%
	Entropy	48.8%	1.5%	13.5%	4.5%	7.4%	46.1%	20.3%
	TTA-VLA (ours)	66.3%	2.7%	26.6%	11.3%	10.7%	62.8%	30.1%
OpenVLA-OFT								
In-distribution	Original Policy			96.7%				96.7%
Out-of-distribution	Original Policy	90.7%	61.0%	98.0%	21.6%	78.9%	75.1%	70.9%
	Entropy	91.9%	62.4%	97.7%	22.6%	79.6%	76.3%	71.8%
	TTA-VLA (ours)	95.5%	71.8%	99.7%	30.0%	87.4%	80.8%	77.5%
π_0								
In-distribution	Original Policy			90.0%				90.0%
Out-of-distribution	Original Policy	79.8%	15.1%	82.5%	5.8%	15.6%	72.6%	45.2%
	Entropy	80.4%	16.0%	83.8%	6.2%	16.5%	73.8%	46.1%
	TTA-VLA (ours)	83.1%	46.2%	88.5%	17.7%	61.0%	75.9%	62.1%



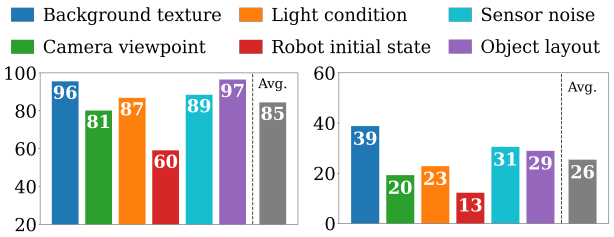
(a) Background (b) Camera viewpoint (c) Light condition (d) Robot initial state (e) Sensor noise (f) Object layout

Fig. 4: Task success rates according to task difficulty under visual perturbations, where difficulty levels on the x-axis follow the original LIBERO-Plus. Blue and orange lines denote the original policy and our TTA-VLA policy, respectively.

adaptation baseline, under the out-of-distribution settings.[2] For each environment, we initialize the model from its publicly available pre-trained checkpoint and follow the original learning hyperparameter settings. For adaptation, we collect a single trajectory using random robot actions within valid action bounds and use three sequentially sampled observations. Using the resulting features, we perform a single adaptation step with a batch size of 1, updating only the visual projector, which accounts for less than 1% of the model parameters. The adapted policy is then used for evaluation.

2) Results: Main results. We summarize the comparison results on six visual perturbation scenarios in TABLE I, reporting success rates averaged over object, spatial, and goal variants of LIBERO-Plus. Across the three VLA backbones, the average success rate decreases from 92.2% in-distribution to 45.1% under visual perturbations, highlighting the sensitivity of pre-trained VLAs to visual distribution shifts. Entropy minimization provides only marginal improvements over the original policies, increasing the average success rate from 45.1% to 46.1%. In contrast, TTA-VLA consistently improves performance across all backbones and perturbation types, achieving an average success rate of 56.6%. The gains are particularly pronounced under camera and sensor perturbations, where TTA-VLA improves the original policies by 14.5% and 19.6% on average, respectively. Overall, these results show

[2]We minimize the categorical entropy of the predicted action-token distributions for OpenVLA, the entropy estimated from stochastic action predictions for π_0, and proxy categorical entropy computed from action representations for OpenVLA-OFT.



(a) Preservation (b) Improvement

Fig. 5: Knowledge preservation and improvement under various visual perturbations compared to the original policy.

that leveraging visual-action transition consistency provides an effective adaptation signal for recovering policy performance under diverse visual perturbations.

Performance according to task difficulty. We analyze our method with respect to task difficulty. In LIBERO-Plus, task instances are grouped into five difficulty levels based on how many evaluated VLAs can complete them, ranging from L1, where all models succeed, to L5, where none succeed. Fig. 4 reports task success rates across difficulty levels for each perturbation type, averaged over OpenVLA and π_0, comparing the original and adapted policies. As expected, performance decreases from L1 to L5 across all perturbations. Nevertheless, our method consistently outperforms the original policy, with average improvements of 16.8% and 10.8% at L1 and L5, respectively. Notably, for background texture, camera

TABLE II: Controlled analyses of TTA-VLA examining (a) the role and task relevance of the language-conditioned anchor, (b) the choice of transition alignment strategy, and (c) the effect of test-time trajectory source.

(a) Language anchor

Anchor	Avg.
No anchor	29.1%
Mismatched	28.3%
Matched	30.1%

(b) Alignment

Alignment	Avg.
Angular	26.5%
Projection	30.1%

(c) Trajectory source

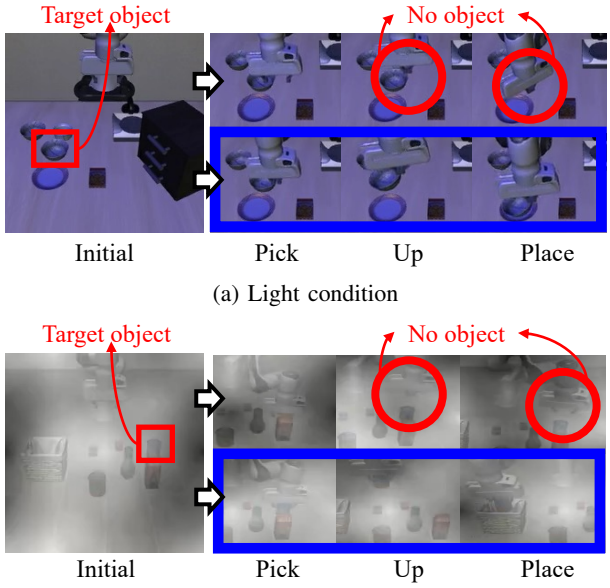
Trajectory	Avg.
Random	30.1%
Policy	30.2%
Demo.	30.5%

viewpoint, and object layout, the performance gap widens at higher difficulty levels (L3-L5), demonstrating its effectiveness on challenging task instances. These results suggest that TTA-VLA improves robustness across task difficulties under visual distribution shifts by leveraging test-time trajectories to mitigate compounded challenges from both task difficulty and visual perturbations.

Knowledge preservation and improvement. To assess the stability and plasticity of our test-time adaptation framework, we analyze the extent to which successful behaviors of the pre-trained policy are preserved and failed ones are improved. Fig. 5 reports the corresponding results across OpenVLA and π_0. *Preservation* denotes the percentage of originally successful tasks that remain successful after adaptation, while *Improvement* denotes the percentage of originally failed tasks that become successful. Our approach preserves most tasks successfully completed by the original policy, achieving an average preservation of 85%. At the same time, our proposal substantially improves previously failed cases, yielding an average improvement of 26%. These results indicate that our method effectively balances maintaining existing successful behaviors and recovering failures.

Analysis of language-conditioned anchoring. To analyze the role of language conditioning, TABLE II(a) compares language-anchor settings using OpenVLA. We first compare the unanchored variant (No anchor) with the correct task instruction (Matched). Its modest improvement over the no-anchor setting indicates an additional benefit from language conditioning, while visual-action transition alignment remains the primary adaptation signal. To determine whether this benefit is associated with task-relevant language information rather than simply with introducing an arbitrary language, we further compare the Matched with a mismatched instruction sampled from a different suite (Mismatch), while keeping the instruction provided to the VLA unchanged. For Mismatch, we use Goal-suite instructions for Object and Spatial and Object instructions for Goal. The mismatched anchor achieves the lowest performance among the three settings. These comparisons suggest that language conditioning provides a modest auxiliary benefit and that this benefit is associated with the task relevance of the language anchor.

Variations of alignment strategies. We further compare projection-based and angular-based feature alignment in $\mathcal{L}_{\text{align}}$ using OpenVLA. Details of alignment approaches are in Section III-B. As shown in TABLE II(b), the projection-based strategy outperforms the angular-based one across all perturbations, with average gains of 3.6%. This suggests that explicitly suppressing deviated components provides more stable alignment than only constraining angular direction.



(a) Light condition

(b) Sensor noise

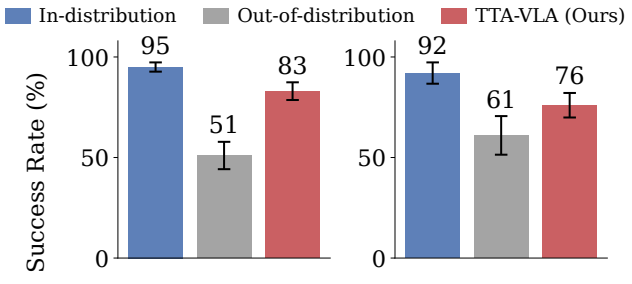
Fig. 6: Visualizations of rollouts with the original policy (top row) and TTA-VLA (bottom row) on LIBERO-Plus under (a) light condition and (b) sensor noise perturbations. Key failure scenes are highlighted with red circles, while successful executions are highlighted with blue boxes.

Effect of trajectory source. To examine whether TTA-VLA depends on the quality of the test-time trajectory, we evaluate three trajectory sources for adaptation: random robot actions (Random), the pretrained policy in the target environment (Policy), and successful source-domain trajectories replayed in the target environment (Demo.), using OpenVLA. As in TABLE II(c), adaptation performance remains comparable across the three trajectory sources, with only minor variations. This suggests that the proposed self-supervised signal does not require successful task execution and does not simply reinforce erroneous behavior.

Visualizations of rollouts. Fig. 6 presents comparisons of rollouts between the original policy and TTA-VLA under light condition and sensor noise perturbations. Under light condition, the original policy exhibits inaccurate grasp poses and fails to stably pick up the target bowl, whereas TTA-VLA produces accurate grasps and completes the task despite illumination changes. Similarly, under sensor noise, the original policy fails to localize the target object, while TTA-VLA correctly identifies it and completes the pick-and-place sequence. More rollout visualizations under other perturbation types are provided in the multimedia materials.

B. Real-world experiments

1) Setup: Tasks. To evaluate robustness while isolating the effects of visual perturbations and adaptation from other factors affecting task performance, we use four fundamental tabletop manipulation tasks representing core manipulation



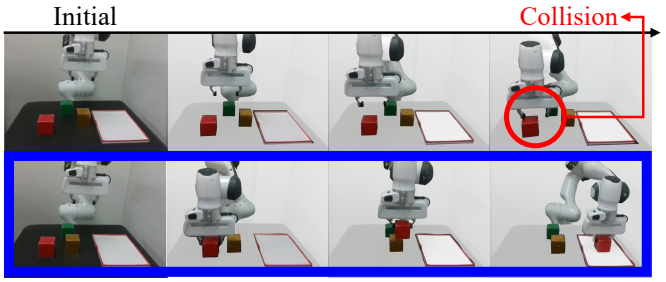
(a) OpenVLA (b) π_0

Fig. 7: Real-world success rates under in-distribution and out-of-distribution settings. For each backbone, results are averaged over the evaluated real-world settings, and error bars denote standard errors.

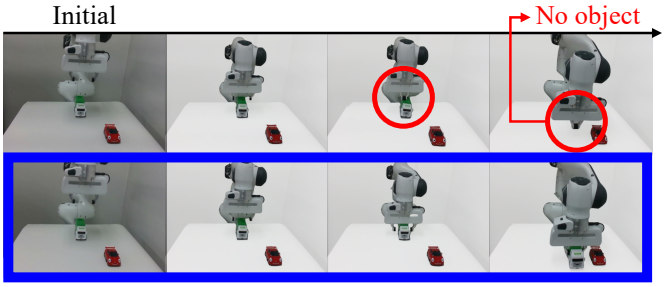
capabilities: pick-and-place, slide, push, and pull. The first task, *Pick up the red block and place it on the plate* (Pick-and-place), requires the robot to localize the target object, grasp it stably, and place it at the designated location. The second task, *Slide the green truck to the red car* (Slide), expects the robot to maintain contact with the object and slide it along the surface toward the target. The third task, *Push the blue button* (Push), requires the robot to position its end effector and apply a directional pushing motion. The fourth task, *Pull out the tissue* (Pull), expects the robot to grasp the tissue and pull it out smoothly without losing contact.

Implementation details. We conduct real-world experiments using a 7-DoF robotic arm Franka Research 3 equipped with a two-finger parallel gripper. For the evaluation, we employ OpenVLA using a frontal third-person camera and π_0 using both third-person and wrist-camera observations, with the corresponding language task instructions. We design four types of visual perturbations, including background texture, camera viewpoint, robot initial state, and object layout. The in-distribution setting uses a black background, a frontal third-person camera view, the home robot pose, and the task-relevant objects at their original locations, respectively. The corresponding out-of-distribution settings are constructed using backgrounds with different colors, alternative camera viewpoints and robot initial poses, and changes in surrounding objects or their locations. π_0 evaluation excludes the Pull task and camera perturbations, limiting direct performance comparison between the two models. We conduct five evaluation trials under different perturbation settings and report the mean performance and standard error across trials.

2) Results: Quantitative results. Fig. 7 summarizes the aggregate real-world performance of OpenVLA and π_0. For OpenVLA, visual perturbations substantially reduce the average success rate from 95% to 51%, while TTA-VLA recovers the performance to 83%. A similar trend is observed for π_0, where TTA-VLA improves the average success rate from 61% to 76% under visual perturbations. The improvements are observed consistently across the evaluated perturbation conditions, despite differences in backbone architecture and action prediction. These results demonstrate that transition-based adaptation remains effective under real-world visual distribution shifts across different VLA backbones.



(a) Object in Pick-and-place



(b) Background in Slide

Fig. 8: Real-world rollouts between the original policy (first row) and our method (second row). Key failure scenes are highlighted with red circles, while successful executions are highlighted with blue boxes.

Visualizations of rollouts. Fig. 8 shows representative real-world rollouts of the original policy (top) and TTA-VLA (bottom). The original policy exhibits unstable behaviors, such as mislocalizing the target object under background shifts in Slide and being distracted by changes in surrounding objects in Pick-and-place, which leads to missing the target or interacting with the wrong region. In contrast, the proposed method maintains stable, target-centric motion and completes the intended manipulation by reducing visual–action inconsistencies induced by visual distribution shifts at the representation-level. These results support that transition-based alignment remains effective under real-world distribution shifts by reducing visual-action inconsistencies. More rollouts are provided in the multimedia materials.

V. Conclusion

We have addressed the challenge of deploying vision–language–action models in environments with unseen visual conditions. To this end, we have introduced TTA-VLA, a test-time trajectory-based adaptation framework that leverages visual–action transition consistency conditioned on language as a self-supervised signal, enabling VLAs to adapt using a small number of test-time observations. To further preserve semantic stability, it anchors transition alignment to the natural-language task instruction, providing an environment-invariant semantic reference that mitigates representational drift. Through extensive experiments on the benchmark and real-world scenarios, we demonstrate that our method significantly improves success rates under diverse visual distribution

shifts. These results highlight that test-time adaptation, driven by simple yet effective self-supervised signals, is a promising direction for enhancing the robustness of VLAs in reliable deployment.

REFERENCES

[1] J. Bjorck, F. Castañeda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu, S. Huang, *et al.*, "Gr00t n1: An open foundation model for generalist humanoid robots," *arXiv preprint arXiv:2503.14734*, 2025.

[2] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, *et al.*, "π_0: A vision-language-action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024.

[3] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, *et al.*, "Openvla: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.

[4] G. R. Team, S. Abeyruwan, J. Ainslie, J.-B. Alayrac, M. G. Arenas, T. Armstrong, A. Balakrishna, R. Baruch, M. Bauza, M. Blokzijl, *et al.*, "Gemini robotics: Bringing ai into the physical world," *arXiv preprint arXiv:2503.20020*, 2025.

[5] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, *et al.*, "Droid: A large-scale in-the-wild robot manipulation dataset," *arXiv preprint arXiv:2403.12945*, 2024.

[6] A. O'Neill, A. Rehman, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain, *et al.*, "Open x-embodiment: Robotic learning datasets and rt-x models," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 6892–6903.

[7] H. R. Walke, K. Black, T. Z. Zhao, Q. Vuong, C. Zheng, P. Hansen-Estruch, A. W. He, V. Myers, M. J. Kim, M. Du, *et al.*, "Bridgedata v2: A dataset for robot learning at scale," in *Conference on Robot Learning*. PMLR, 2023, pp. 1723–1736.

[8] Y. Ma, Z. Song, Y. Zhuang, J. Hao, and I. King, "A survey on vision-language-action models for embodied ai," *arXiv preprint arXiv:2405.14093*, 2024.

[9] R. Sapkota, Y. Cao, K. I. Roumeliotis, and M. Karkee, "Vision-language-action models: Concepts, progress, applications and challenges," *arXiv preprint arXiv:2505.04769*, 2025.

[10] S. Fei, S. Wang, J. Shi, Z. Dai, J. Cai, P. Qian, L. Ji, X. He, S. Zhang, Z. Fei, *et al.*, "Libero-plus: In-depth robustness analysis of vision-language-action models," *arXiv preprint arXiv:2510.13626*, 2025.

[11] G.-C. Kang, J. Kim, K. Shim, J. K. Lee, and B.-T. Zhang, "Clip-rt: Learning language-conditioned robotic policies from natural language supervision," *arXiv preprint arXiv:2411.00508*, 2024.

[12] Y. Hu, F. Lin, P. Sheng, C. Wen, J. You, and Y. Gao, "Data scaling laws in imitation learning for robotic manipulation," *arXiv preprint arXiv:2410.18647*, 2024.

[13] Y. Ji, H. Tan, J. Shi, X. Hao, Y. Zhang, H. Zhang, P. Wang, M. Zhao, Y. Mu, P. An, *et al.*, "Robobrain: A unified brain model for robotic manipulation from abstract to concrete," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 1724–1734.

[14] Y. Yang, J. Sun, S. Kou, Y. Wang, and Z. Deng, "Lohovla: A unified vision-language-action model for long-horizon embodied tasks," *arXiv preprint arXiv:2506.00411*, 2025.

[15] M. Zawalski, W. Chen, K. Pertsch, O. Mees, C. Finn, and S. Levine, "Robotic control via embodied chain-of-thought reasoning," *arXiv preprint arXiv:2407.08693*, 2024.

[16] S. Kim, G. Oh, H. Ko, D. Ji, D. Lee, B.-J. Lee, S. Jang, and S. Kim, "Test-time adaptation for online vision-language navigation with feedback-based reinforcement learning," in *Forty-second International Conference on Machine Learning*, 2025.

[17] M. Shu, W. Nie, D.-A. Huang, Z. Yu, T. Goldstein, A. Anandkumar, and C. Xiao, "Test-time prompt tuning for zero-shot generalization in vision-language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 14 274–14 289, 2022.

[18] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," *arXiv preprint arXiv:2006.10726*, 2020.

[19] C.-M. Feng, K. Yu, Y. Liu, S. Khan, and W. Zuo, "Diverse data augmentation with diffusions for effective test-time prompt tuning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2704–2714.

[20] Y. Liu, P. Kothari, B. Van Delft, B. Bellot-Gurlet, T. Mordan, and A. Alahi, "Ttt++: When does self-supervised test-time training fail or thrive?" *Advances in Neural Information Processing Systems*, vol. 34, pp. 21 808–21 820, 2021.

[21] C. K. Mummadi, R. Hutmacher, K. Rambach, E. Levinkov, T. Brox, and J. H. Metzen, "Test-time adaptation to distribution shift by confidence maximization and input transformation," *arXiv preprint arXiv:2106.14999*, 2021.

[22] Z. Nado, S. Padhy, D. Sculley, A. D'Amour, B. Lakshminarayanan, and J. Snoek, "Evaluating prediction-time batch normalization for robustness under covariate shift," *arXiv preprint arXiv:2006.10963*, 2020.

[23] J. Gao, X. Yao, and C. Xu, "Fast-slow test-time adaptation for online vision-and-language navigation," *arXiv preprint arXiv:2311.13209*, 2023.

[24] M. J. Kim, C. Finn, and P. Liang, "Fine-tuning vision-language-action models: Optimizing speed and success," *arXiv preprint arXiv:2502.19645*, 2025.

[25] L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, D. Salz, M. Neumann, I. Alabdulmohsin, M. Tschannen, E. Bugliarello, *et al.*, "Paligemma: A versatile 3b vlm for transfer," *arXiv preprint arXiv:2407.07726*, 2024.

[26] S. Karamcheti, S. Nair, A. Balakrishna, P. Liang, T. Kollar, and D. Sadigh, "Prismatic vlms: Investigating the design space of visually-conditioned language models," in *Forty-first International Conference on Machine Learning*, 2024.

[27] K. Pertsch, K. Stachowicz, B. Ichter, D. Driess, S. Nair, Q. Vuong, O. Mees, C. Finn, and S. Levine, "Fast: Efficient action tokenization for vision-language-action models," *arXiv preprint arXiv:2501.09747*, 2025.

[28] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Conference on Robot Learning*. PMLR, 2023, pp. 2165–2183.

[29] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "Rdt-1b: a diffusion foundation model for bimanual manipulation," *arXiv preprint arXiv:2410.07864*, 2024.

[30] J. Wen, Y. Zhu, J. Li, M. Zhu, Z. Tang, K. Wu, Z. Xu, N. Liu, R. Cheng, C. Shen, *et al.*, "Tinyvla: Towards fast, data-efficient vision-language-action models for robotic manipulation," *IEEE Robotics and Automation Letters*, 2025.

[31] J. Liang, R. He, and T. Tan, "A comprehensive survey on test-time adaptation under distribution shifts," *International Journal of Computer Vision*, vol. 133, no. 1, pp. 31–64, 2025.

[32] Z. Xiao and C. G. Snoek, "Beyond model adaptation at test time: A survey," *arXiv preprint arXiv:2411.03687*, 2024.

[33] S. Schneider, E. Rusak, L. Eck, O. Bringmann, W. Brendel, and M. Bethge, "Improving robustness against common corruptions by covariate shift adaptation," *Advances in neural information processing systems*, vol. 33, pp. 11 539–11 551, 2020.

[34] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, and M. Hardt, "Test-time training with self-supervision for generalization under distribution shifts," in *International conference on machine learning*. PMLR, 2020, pp. 9229–9248.

[35] J. Kwok, C. Agia, R. Sinha, M. Foutter, S. Li, I. Stoica, A. Mirhoseini, and M. Pavone, "Robomonkey: Scaling test-time sampling and verification for vision-language-action models," in *Proceedings of The 9th Conference on Robot Learning*, 2025.

[36] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.

[37] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proceedings of the IEEE international conference on computer vision*, 2013, pp. 2960–2967.

[38] J. Cha, W. Kang, J. Mun, and B. Roh, "Honeybee: Locality-enhanced projector for multimodal llm," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 13 817–13 827.

[39] Y. Zhai, S. Tong, X. Li, M. Cai, Q. Qu, Y. J. Lee, and Y. Ma, "Investigating the catastrophic forgetting in multimodal large language models," *arXiv preprint arXiv:2309.10313*, 2023.

[40] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "Libero: Benchmarking knowledge transfer for lifelong robot learning," *Advances in Neural Information Processing Systems*, vol. 36, pp. 44 776–44 791, 2023.